

# DINO-4DSTEM

A label-free tool for reading 4D-STEM diffraction maps

User manual

For first-time and occasional users. No programming needed.

This program takes a 4D-STEM dataset and automatically sorts every scan position into a small number of “classes” that share the same diffraction, giving you a colour map of your sample, without labels, without choosing the number of classes, and without setting peak thresholds. It can also run the classical comparison methods (NMF, ACOM, SAM) and tell you what the classes mean, all in one window.

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## 1. What this program does (in plain words)

A 4D-STEM experiment scans a tiny electron beam across your sample and, at every scan position, records a 2D *diffraction pattern*. The result is a large 4-dimensional dataset: a 2D grid of positions, each holding a 2D diffraction image. Looking through thousands of these patterns by hand is slow and subjective.

DINO-4DSTEM learns, on its own, to group scan positions whose diffraction looks alike. The output is a *class map*: an image of your scan where each colour is one group (“class”) of similar diffraction. From there you can look at the average diffraction of each class, ask the program what physically distinguishes the classes, and compare the result against the traditional methods.

**Three things make it easy to use:**

- **No labelling.** You do not have to tell it what to look for.
- **No “number of classes” to guess.** It finds how many are needed.
- **No thresholds to tune.** It does not rely on peak-finding settings that you would otherwise have to adjust by trial and error.

**Note.** You can use the program in two ways. (1) *Train* a fresh model on your own data (takes a while, uses the graphics card), or (2) *load an existing trained model* and just look at and analyse the results (fast). Both are covered below.

## 2. Starting the program and the main window

### 2.1 Starting it

Launch the program the way it was set up on your computer (a desktop shortcut, or by running `gui_dino4dstem.py` in the project’s Python environment). A single window titled **DINO-4DSTEM** opens. A small black text window may also appear behind it; that is the program’s log. You can ignore it, but do not close it, as closing it closes the program.

**Tip.** Maximise the window (the square button at the top-right) before you start, so all the controls and images are visible.

### 2.2 The top bar (always visible)

Across the very top of the window, from left to right:

- **Inspect runs (HTML)** : opens a summary page of all the models you have trained, in your web browser. Useful for finding an earlier result.
- **Assistant** : opens a plain-English chat helper that can carry out common actions for you by typing requests in everyday language. (Optional; may not be present in every version.)
- **dataset badge** (“dataset: click to load a cube”): shows which data file is currently loaded. Click it to load one.
- **run badge** (“run: none”): shows which trained model is currently in use. Click it to pick one.
- **inference dot** : a small status light that tells you whether a class map has been computed yet.

- **real space** (nm/px): the distance between neighbouring scan positions, i.e. your scan step size in nanometres. Used only for the scale bar and axis labels on real-space images.
- **reciprocal** (nm<sup>-1</sup>/px): the calibration of the detector, in nm<sup>-1</sup> per original detector pixel, as reported by your microscope for the camera length you used. Used for the diffraction scale bars and the 1D plots.

**Note.** You only ever type these two calibration numbers once. Every plot in the program automatically applies the right scaling, so you do not need to recompute anything when the program resizes images internally.

## 2.3 How the tabs are organised

The work is grouped into five big tabs along the top:

- **Data** : load your 4D-STEM file and prepare it.
- **Model** : train a model, watch it learn, and apply it to other data.
- **Analysis** : look at the class map and understand what the classes mean.
- **Clustering** : the classical comparison methods (NMF, pretrained-DINO, SAM).
- **Diffraction** : per-pattern tools (peak/blob detection and crystal orientation mapping, ACOM).

Each big tab contains smaller sub-tabs (for example **Data** contains **Load + Pre-process** and **Synthetic**). Click a big tab first, then the sub-tab you want.

## 3. The five-minute workflow

If you just want a class map of your data, here is the whole path. Each step is explained in detail later.

1. **Data → Load + Pre-process**: click **Browse**, choose your 4D-STEM file, click **Load**. Set **vmax** so the rings are visible without the centre being a white blob. Leave the crop/mask at their defaults unless you have a reason to change them.
2. Enter your **real space** and **reciprocal** calibration in the top bar.
3. **Model → Training**: pick your **Sample**, leave **Use paper recipe (locked)** on, and click **Train**. Wait for it to finish. *Or*, to skip training, click the **run badge** and load an existing trained model.
4. **Model → Eval** (or **Analysis → Post-hoc**): view the class map.
5. **Analysis → Post-hoc → Interpretation**: click **Run interpretation** to get figures and a written report on what the classes mean.

## 4. Data: loading and preparing your file

### Data → Pre-processing → Load + Pre-process

This is where you open your data and get it ready. It is the first tab you use.

## 4.1 Loading a file

- **Browse then Load:** open a single 4D-STEM cube. Supported file types include `.prz`, `.npy`, `.npz`, and `.h5`.
- **Browse multi...** : open several files together (for example four quadrants of one acquisition).

After loading, the left side shows one diffraction pattern, and the bottom says how the data is shaped. If nothing appears, check the log window for a message.

## 4.2 Looking through the patterns

- **Pattern index** slider: moves through the scan positions. Click the slider or the image first, then use the left/right arrow keys for single steps, or Page-Up/Page-Down for jumps of ten. **Random** jumps to a random position.
- **vmax**: the brightness ceiling. Lower it until you can see the faint diffraction rings; raise it if the image is too dark. This only changes the display and how the model sees intensities, not your raw file.
- **cmap**: the colour scheme of the diffraction image.
- **log-stretch**: compresses very bright and very faint together; helpful when the central beam is far brighter than the rings.

**Tip.** Choosing a good **vmax** is the single most useful thing you do here. Aim for a picture where both the bright centre and the outer rings are visible.

## 4.3 Crop and central-beam mask

These define what part of each diffraction pattern the model looks at.

- **center crop (px, in 192 frame)**: how far out from the centre the model looks. Keep your rings of interest comfortably inside this circle.
- **polar mask cols (low-r mask in polar)**: hides the very centre (the direct beam) so the intense central spot does not dominate. Too small lets the beam bleed in; too large eats the first ring.

**Note.** The defaults are sensible for most molecular-film data. Only change them if you know the ring you care about is being cut off (too-small crop) or the centre is swamping everything (too-small mask).

## 4.4 Other preparation tools (optional)

- **Binning (down-sample)**: **bin factor n** then **Apply binning** writes a new, smaller, faster file. Use only if loading is slow. Choose **real-space** binning to merge scan positions or **q-space** to merge detector pixels.
- **Gaussian blur**:  $\sigma$  (px) then **Apply blur to dataset** writes a gently smoothed copy. Reduces speckle noise on very low-dose data.
- **Ellipticity correction**: fixes a slightly oval detector. Use **Auto-fit**, or type **a/b** and  $\theta$  and click **Apply**. **Reset** returns to no correction.

- **NBED centering: COM-center the direct beam / Center entire dataset** re-centres each pattern on the beam if your data drifts.
- **Pair labelling (pre-train)**: an advanced option to give the model a few “these two look the same / different” hints. Most users never need this.

**Caution.** The buttons that say “write new .cube.npy” create a new file on disk; they do not overwrite your original. Note the new file name so you load the right one for training.

## 4.5 Synthetic (advanced)

**Data → Synthetic** simulates diffraction data from a known crystal structure, to test the method. It is for advanced users and is not needed for normal analysis.

# 5. Model: training and applying a model

## 5.1 Training

**Model → Training → Training**

This teaches a model on your loaded data.

1. Choose the **Sample** (the data you prepared).
  2. Optionally type a **Run name** so you can find the result later.
  3. Leave **Use paper recipe (locked)** switched on. This applies the validated default settings, so you do not have to choose anything.
  4. Click **Train**. Progress shows in the status line; you can watch the map form live on the **Eval** sub-tab. Click **Stop** to end early.
- **Refresh crop/mask/COM from PrePanel**: copies the crop and mask you set on the Data tab into the training settings, so the model trains on exactly what you previewed.
  - To change advanced settings, click the lock switch to unlock the recipe, then **Apply changes**. **Apply auto-default** ( $\lambda_{\text{pair}} = 0.1$ ) turns on the optional pair-hint term if you provided pair labels.

**Caution.** Training uses the graphics card and can take a while. Avoid running other heavy GPU work at the same time. If your data loading feels stuck, that is usually disk speed, not a crash; give it time.

## 5.2 Eval: viewing the class map

**Model → Training → Eval**

- During training this updates automatically (“awaiting first checkpoint...” then the live map).
- To view a finished model: **Load run dir...** (**model**), pick the run folder, and click **Render**. **Browse cube...** lets you apply the model to a different data file.

### 5.3 Transfer: applying one model to many files

#### Model → Transfer

Use a trained model on a whole folder of other datasets in one go.

1. Point **Cubes folder** at the folder, click **Refresh**, and tick the files you want (up to twenty).
2. Pick the starting model under **ckpt**.
3. Click **Run sequential transfer**. **Stop** ends it.
  - **freeze encoder (head-only adapt)**: a faster, lighter adaptation.
  - **Chain ckpts**: each file starts from the previous file's result; useful for a time series.

## 6. Analysis: understanding the class map

### 6.1 Post-hoc: explore the map

#### Analysis → Post-hoc → Analysis

This is the main place to look at results.

- **Load run dir...** and **Browse cube...**: load a trained model and its data (or use the ones already in the session).
- **Click the class map** to inspect a position: it shows that position's raw pattern. Clicking a class shows the **class average** diffraction ("class avg (top 200)"), which is much cleaner than a single noisy frame.
- **Attribution (Avg + GradCAM + IG)**: **Compute GradCAM + IG** makes a picture showing *which part of the diffraction the model used* to decide a class. Choose the **CAM layer** if prompted; the defaults are fine.
- **Merge classes**: tick two or more classes and click **Apply merge** to combine ones you consider the same. **Reset** undoes it.
- **Fine-tune with labels**: an advanced option to nudge the model using a few of your own pair labels.
- **Send grain to ACOM tab**: hands the selected region to the orientation- mapping tool (see Diffraction).
- **Saving**: **Save snapshot**, **Save panel (PDF+PNG)**, and **Report All (PPTX + PNGs)** export your figures.

**Tip.** The class average is your friend. A single low-dose pattern looks like noise; the class average reveals the real rings and spots.

### 6.2 Interpretation: what do the classes mean?

#### Analysis → Post-hoc → Interpretation

This runs a battery of checks and writes a short report.

1. Tick the **Analyses to run** you want (leave all on for a full report).
2. Click **Run interpretation**. When done, use the **Figure** selector to flip through the results, or **Open output folder** to find the files.
3. **Refresh from Post-hoc** re-reads the currently loaded run.

The report answers, in figures and text: which physical quantities the classes track (for example crystallinity or texture), whether the splits are real or redundant, and how the model compares to the classical methods.

## 7. Clustering: the classical comparison methods

These tabs let you run the traditional approaches on the *same* data, so you can compare them with the model directly.

### 7.1 NMF + cluster

#### Clustering → NMF + cluster

Non-negative matrix factorization followed by clustering (a standard baseline).

1. **Load .prz / .npz** (or use the session sample).
2. Set **n.comp** (how many components) and pick the **method** (**K-means** is the default; **Agglomerative**, **FCM**, **HDBSCAN** are alternatives). **K** is the number of clusters, or choose **auto**.
3. Click **Run**. **Show class averages** and **Open interactive map** display the result; **Save PNG / Report All** export it.

### 7.2 DINO + cluster

#### Clustering → DINO + cluster

Uses a general, pre-trained image model (downloaded automatically) instead of one trained on your data, then clusters its features. A second baseline. Controls mirror the NMF tab, plus **Show UMAP** (a 2D scatter of the patterns) and **Class attentions (grid)**.

### 7.3 SAM

#### Clustering → SAM

Uses the “Segment Anything” model to outline features in a class-average diffraction pattern.

- Set **SAM ckpt** to the model file, pick a **class**, and click **Run on this average**.
- **Apply to whole class / Apply to ALL classes** repeat it broadly.
- **Save tuning to per-class config** remembers your settings per class.

## 8. Diffraction: per-pattern tools



## 8.1 Blob (peak detection)

### Diffraction → Blob

Finds spots/peaks in a diffraction image.

- Pick a **class** and click **Run on current image**, or set a **frame idx / grain (y, x)** to target a specific pattern.
- **Method** chooses the detector; **log1p stretch** helps faint peaks show up. **Apply to whole class / Apply to ALL classes** run broadly.

## 8.2 ACOM (crystal orientation mapping)

### Diffraction → ACOM

Matches each pattern to a known crystal structure (from a CIF file) to find crystal *orientation*. It works in clear steps, left to right:

1. **Load source**: pulls the current data and class map from the session (no separate load button; it uses the top-bar badges).
2. **+ CIF...** then **Build all**: add your structure file(s) and build the reference reflections.
3. **Detect peaks**: finds peaks in the patterns. **log stretch before detection**, **Top-N**, and **skip if ; N peaks** control sensitivity.
4. **Calibration**: set **1/Å per px** (or click **Auto-fit**) and **k\_max**.
5. **Match single**: solve one position (type a scan (y, x) or click the class map). **Single-phase full / Multi-phase full** solve the whole scan (**full-dataset stride** trades speed for completeness; **corr threshold** and **winner margin** control acceptance).
6. **Overlay** shows the orientation map; tick **also show in-plane orientation** for the rotation angle.

**Note.** ACOM needs a structure file (CIF) and a few detection settings, which is exactly the kind of manual tuning the model avoids. Use it as a cross-check or when you specifically want crystal orientation, which the model deliberately ignores.

## 9. Where your results are saved

- Trained models live in a **runs** folder, one folder per model, each holding the model file, a settings file, a training log, and an **eval** folder with the class map.
- Figures and reports you export (snapshots, panels, “Report All”) are written next to the run or to a folder the button names; use **Open output folder** on the Interpretation tab to find them.
- New data files you create (binned, blurred, centred) are written as new **.cube.npy** files in the same place as the original.

## 10. Common tasks at a glance

- **Just show me a class map of my data, using an existing model.** Click the **run badge** (top bar) and load the model; click the **dataset badge** and load your file; go to **Model → Eval** and click **Render**.
- **Train a new model on my data.** **Data** load + set **vmax**; **Model → Training**, pick the sample, keep the locked recipe, **Train**.
- **Understand what the classes are.** **Analysis → Post-hoc → Interpretation**, **Run interpretation**.
- **Compare to the classical method.** **Clustering → NMF + cluster**, **Run**, then compare its map to the model's.
- **Get crystal orientations.** **Diffraction → ACOM**, add a CIF, **Build all**, **Detect peaks**, **Single-phase full**, **Overlay**.
- **Apply one model to many files.** **Model → Transfer**, pick the folder, tick files, **Run sequential transfer**.

## 11. Troubleshooting

- **Nothing happens after I click Load.** Check the small black log window for a message; the file type or scan shape may not have been recognised. Try **Browse** again and confirm the file.
- **The diffraction image is all white in the middle, or all black.** Adjust **vmax** on the Data tab and turn on **log-stretch**.
- **The class map looks empty or a single colour.** Make sure a model is loaded (run badge) and a class map has been computed (the inference light); on the Eval tab click **Render**.
- **Training seems frozen.** Low-dose data loads slowly; wait a few minutes. If the graphics card is busy with other work, training will be slow.
- **A figure looks cramped or mislabelled.** Resize the window larger and re-render; many panels redraw to fit.
- **I changed crop/mask but training ignored it.** On the Training tab click **Refresh crop/mask/COM** from **PrePanel** before **Train**.

## 12. Glossary

### 4D-STEM cube

The whole dataset: a 2D grid of scan positions, each with a 2D diffraction pattern.

### Diffraction pattern

The 2D image recorded at one scan position; rings and spots in it encode the local structure.

**Class / class map**

A group of scan positions with similar diffraction, and the colour image showing where each group sits on the sample.

**Class average**

The averaged diffraction of all positions in a class; cleaner than any single pattern.

**Training**

Letting the model learn groupings from your data. Done once per dataset; afterwards the model can be reloaded instantly.

**Run**

One trained model and its outputs, stored in its own folder.

**vmax**

The brightness ceiling used when showing and reading diffraction intensities.

**Crop / mask**

How much of each pattern the model looks at, and how much of the bright central beam is hidden.

**NMF, SAM, ACOM**

Classical analysis methods provided for comparison: decomposition-plus-clustering, feature segmentation, and crystal-orientation mapping, respectively.

**Calibration (real / reciprocal)**

The two numbers, scan step size (nm/px) and detector calibration ( $\text{nm}^{-1}/\text{px}$ ), that set the scale bars and axes.

**Getting help.** If you are unsure which setting to use for a new material, the program's defaults (the locked "paper recipe") are a safe starting point. The on-screen "?" help buttons next to many controls explain each one in place.